

Dear Analog Team,

I am a third-year Computer Science and Information Technology student at the Technical University of Cluj-Napoca, with a strong interest in embedded systems and digital hardware design. I am writing to express my interest in the Analog Devices Summer Practice Program.

Throughout my academic journey, I have focused on understanding how systems operate at a low level, particularly at the intersection of hardware and software. This interest is reflected in my hands-on projects, where I worked extensively with VHDL, FPGA platforms, and embedded systems.

One of my most relevant projects is a Zynq SoC hardware–software accelerator, where I designed a custom VHDL IP core for fixed-point computations and integrated it using AXI communication. I also developed the software side in C, enabling communication and control through a UART interface. This project gave me valuable experience in hardware–software co-design, system integration, and validation.

In addition, I developed an FPGA-based serial communication protocol using VHDL, where I designed FSM-based architectures and verified functionality through simulation and hardware deployment. I also worked on an embedded system project involving SPI, I2C, UART, and PWM communication, which strengthened my understanding of real-time systems and low-level interfacing.

I am particularly interested in Analog Devices because of its strong focus on high-performance signal processing and hardware innovation. I am motivated to further develop my skills in FPGA design, digital systems, and embedded engineering in a professional environment, and I am eager to learn from experienced engineers while contributing to real-world projects.

I believe I would be a good fit for this program due to my strong technical foundation, practical project experience, and genuine interest in low-level system design. I am highly motivated, detail-oriented, and comfortable working with complex systems.

Thank you for your time and consideration. I would welcome the opportunity to contribute and further develop within your team.

Kind regards,
Ilies Bianca Izabela